Rational Functions: Final Retake Practice: Parent Key

Worked solutions and proposed practice rubric. Keep separate from the student copy.

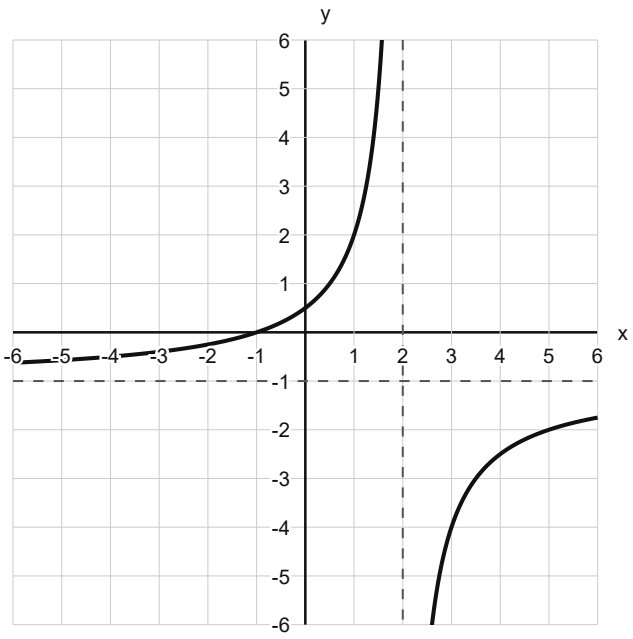
13 questions | 58 points | 75 minutes (user-selected allowance)

Read answers in their printed context and credit valid work wherever it appears. Within dependent work, deduct an originating error once and award mathematically correct follow-through. A separate omission, such as not drawing a calculated hole or not providing requested verification, is independently scorable. Independent errors on different questions are not automatically excused. Accept equivalent valid functions and their own correct hole heights. This is not an official school key.

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1. 8 points

1. Rewrite $f(x) = -\dfrac{x+1}{x-2} = -1 - \dfrac{3}{x-2}$. Thus the vertical asymptote is $x = 2$ and the horizontal asymptote is $y = -1$.
2. For example, $f(1) = 2$, $f(3) = -4$, and $f(0) = 1/2$. Left of 2 the curve lies above -1; right of 2 it lies below -1.
3. As $x \to -\infty$, $f(x) \to -1$; as $x \to +\infty$, $f(x) \to -1$. The side of approach is useful but is not separately required for the end-limit points.



Practice rubric (8 points)

- 4: Graph: correct asymptote equations 1 (0.5 each), two correctly labeled points on different branches 1 (0.5 each), consistent left/right branches 2 (1 each). Do not penalize small freehand waviness as a new conceptual error.
- 2: Far-left: correct approach to -1 in the labeled box or an explicit signed-input statement. 'Approaches -1' in that box earns full credit; a bare correct height earns 1. Side of approach is not additionally required.
- 2: Far-right: correct approach to -1 in the labeled box or an explicit signed-input statement. 'Approaches -1' in that box earns full credit; a bare correct height earns 1. Credit correct statements wherever they appear.

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2. 4 points

1. Factor the denominators as $(x+3)(x-2)$ and $(x+3)(x+2)$. Their common denominator is $(x+3)(x-2)(x+2)$.
2. The numerator is $2(x+2)-3(x-2)=10-x$. No factor cancels. Retain the original exclusions $-3, -2, 2$.

Practice rubric (4 points)

- 1: Correct factorizations and common denominator; 0.5 for correctly factoring both but not completing the LCD.
- 2: Correct signed subtraction: 1 for setup 2(x+2)-3(x-2), 1 for numerator 10-x and complete fraction.
- 1: Original exclusions: award one-third per correct exclusion (-3, -2, 2), subtract one-third per distinct spurious exclusion, and bound this component between 0 and 1. Retain exact thirds until totaling.

3. 4 points

1. Factor as $\dfrac{(x-3)(x+3)}{(x+2)(x+3)} \cdot \dfrac{(x+2)^2}{(x-3)^2}$.
2. Cancel only matching factors to leave $(x+2)/(x-3)$. The original denominator excludes $-3, -2, 3$. In particular -2 remains excluded even though the reduced numerator vanishes there.

Practice rubric (4 points)

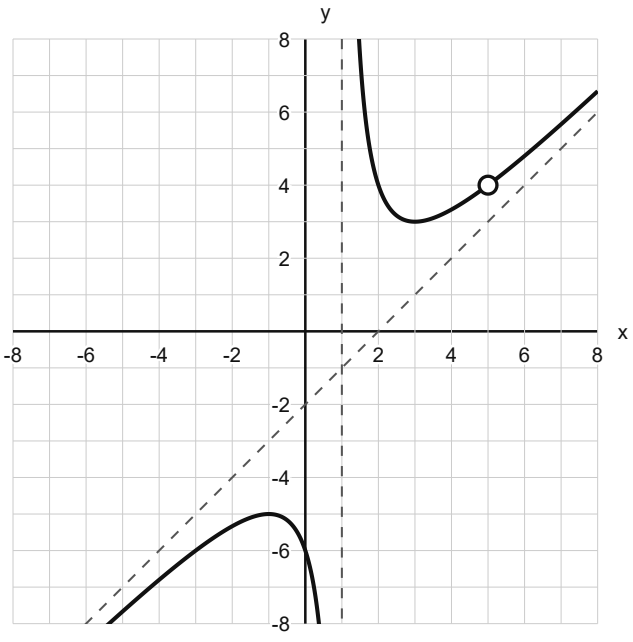
- 1: Correct factorizations, 0.25 for each original polynomial.
- 2: Valid repeated-factor cancellation 1; correct remaining fraction 1.
- 1: Original exclusions: award one-third per correct exclusion (-3, -2, 3), subtract one-third per distinct spurious exclusion, and bound this component between 0 and 1. Retain exact thirds until totaling.

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4. 8 points

1. The parent-only generating rule is $F(x) = x - 2 + 4/(x-1)$, with $x \neq 1, 5$. Students read the graph; no calculus or generating rule is required.
2. The graph increases on $(-\infty, -1) \cup (3, 5) \cup (5, \infty)$, and decreases on $(-1, 1) \cup (1, 3)$. The missing input 5 splits the increasing branch, not its direction.
3. Relative maximum $(-1, -5)$, relative minimum $(3, 3)$. It is concave down on $(-\infty, 1)$ and concave up on $(1, 5) \cup (5, \infty)$.
4. The asymptotes are $x = 1$ and $y = x - 2$, with no horizontal line. A consistently repeated omission of the hole from increasing and concavity intervals is deducted once, not twice.



Practice rubric (8 points)

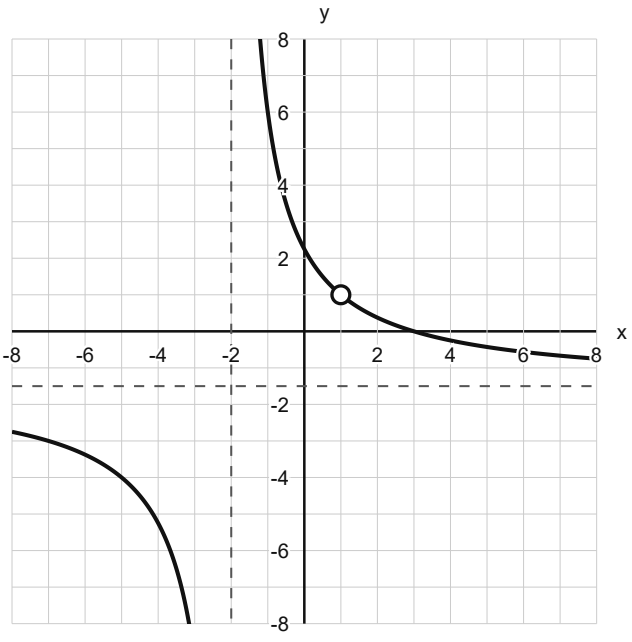
- 1: Increasing: 0.5 correct direction/turning boundaries; 0.5 correct domain partition including the hole at 5. Accept union, 'and', or comma-separated interval lists equally; assess excluded inputs, not the connective.
- 1: Decreasing: 0.5 for each of (-1,1) and (1,3); split at the vertical asymptote. Accept union, 'and', or comma-separated interval lists equally.
- 2: Correct maximum and minimum with type and coordinates, 1 each.
- 2: Concavity: 1 per direction with correct boundary/coverage; retain 0.5 each for correct direction if a boundary is wrong. If the sole error is the same missed hole already deducted in increasing, award that concavity partition credit.
- 2: VA x=1: 0.75; slant y=x-2: 0.75; no HA: 0.5. Credit correctly located lines separately from equation-variable mistakes.

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5. 10 points

1. Factor $R(x) = \dfrac{-3(x-1)(x-3)}{2(x+2)(x-1)} = \dfrac{-3(x-3)}{2(x+2)}$, retaining $x \neq -2, 1$.
2. The intercepts are $(3, 0)$ and $(0, 9/4)$. At the excluded input 1, the reduced rule gives $-3(1-3)/(2(1+2)) = 1$, so the hole is $(1, 1)$.
3. The vertical line is $x = -2$; the horizontal line is $y = -3/2$. Write $R(x) = -3/2 + (15/2)/(x+2)$. There is no slant asymptote.
4. The left branch is below -3/2 and falls to negative infinity near -2. The right branch decreases from positive infinity, passes through the actual intercepts, and approaches -3/2 from above. Draw an open circle at (1,1).



Practice rubric (10 points)

- 2: Factoring/cancellation 1.5; original exclusions -2 and 1: 0.5.
- 2: x-intercept (3,0) and y-intercept (0,9/4), 1 each.
- 2: Hole input 1 and exact height 1, 1 each; shown substitution supports the height.
- 2: VA x=-2 and HA y=-3/2, 1 each. For each, retain 0.5 if the correct location is demonstrated but its equation uses the wrong variable.
- 2: Sketch: consistent branches 1 (0.5 each), correctly located open hole 0.5, labeled features 0.5. Preserve follow-through and do not repeat a copied feature/equation error.

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6. 4 points

1. Division gives $S(x) = x/2 + 3/2 - 4/(2x - 2)$. Check: $(x/2 + 3/2)(2x - 2) = x^2 + 2x - 3$, leaving remainder -4.
2. Thus the slant is $y = x/2 + 3/2$, and no horizontal asymptote exists. As $x \to -\infty$, $S(x) \to -\infty$; as $x \to +\infty$, $S(x) \to +\infty$.
3. The difference $S(x) - (x/2 + 3/2) = -4/(2x - 2)$ tends to zero at both ends. It is not zero at any finite allowed input.
4. A comparison with the slant alone does not complete the requested output statement. Credit correct explicit statements wherever they appear; reversed box placement alone is not a mathematical deduction. These final-rehearsal requirements do not revise previous practice awards.

Practice rubric (4 points)

- 1.5: Correct quotient 1 and remainder -4/exact identity 0.5.
- 1: Correct slant line 0.75 and no-horizontal conclusion 0.25.
- 0.5: Correct far-left output S(x) tending to negative infinity, or equivalent words. The labeled box supplies the input context.
- 0.5: Correct far-right output S(x) tending to positive infinity, or equivalent words. The labeled box supplies the input context.
- 0.5: Explains that the remainder fraction tends to zero at the far ends. Award 0.25 for the correct fraction written equal to zero without a limiting explanation; calculating the remainder alone earns no explanation credit.

7. 4 points

1. Rewrite $C(t) = 3 - 5/(5 + 2t)$. For every finite $t \geq 0$, the subtracted fraction is strictly positive.
2. The concentration approaches 3 grams per liter from below as more solution enters. Since $5/(5 + 2t) \neq 0$ at finite allowed times, it never equals 3 then.

Practice rubric (4 points)

- 1: Correct limit 3.
- 1: Correct grams-per-liter interpretation in context.
- 1: Correct approach from below, supported by signed algebra or equivalent valid reasoning.
- 1: Correct nonattainment conclusion with valid finite-time justification; half if conclusion alone.

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8. 4 points

1. One valid rule is $G(x) = \dfrac{-2x(x+2)}{x(x-3)}$, with $x \neq 0, 3$. Its reduced rule is $-2(x+2)/(x-3)$.
2. The remaining numerator vanishes only at -2, the remaining denominator only at 3, the leading-coefficient ratio is -2, and the canceled factor creates the only hole at 0. These statements, or equivalent labeled algebra, verify the features.
3. For this rule the missing height is $-2(0+2)/(0-3) = 4/3$. There is no y-intercept because the original function is undefined at 0, even though its reduced rule has a finite value there. Accept any rule meeting all conditions and its own correct hole height.
4. Score construction conditions independently. Award correct verification and actual hole-coordinate follow-through for the submitted rule even if another feature is wrong. A rule alone does not earn the separate verification point. Do not invent a hole where the submitted rule has none.

Practice rubric (4 points)

- 2: Valid factored construction: only required zero, pole, horizontal height and hole, 0.5 each.
- 1: Verification: correct original exclusions with cancellation/domain distinction 0.5; explicit correct checks of remaining zero/pole and signed leading ratio 0.5. Labeled algebra counts, prose is not required. Credit correct checks of the submitted rule even if construction misses another requested feature.
- 0.5: Correct actual hole coordinates for the submitted rule, with shown substitution, even if another required feature is wrong; retain 0.25 for correct input if height is wrong.
- 0.5: No y-intercept with the original-domain explanation; 0.25 for conclusion alone.

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9. 2 points

1. Both ends approach the horizontal line $y = -2$, which is finite and nonzero. Therefore numerator and denominator degrees are equal.
2. A smaller numerator degree would give end height 0; a larger numerator degree would not give this finite horizontal end height. Exact individual degrees are not determined from the picture.

Practice rubric (2 points)

- 1: Choice C.
- 1: Connects the finite nonzero horizontal height to equal degrees; 0.5 for identifying the correct horizontal height without the degree connection.

10. 2 points

1. The double zero at 1 makes the curve touch the x-axis rather than cross. D has vertical asymptotes at $x = -2, 2$, horizontal asymptote $y = 1$, and negative values between the vertical asymptotes except at the zero.
2. A crosses at 1 and has horizontal height 0. B touches but has opposite branch signs and horizontal height -1. C has no real vertical asymptotes and has horizontal height 1/2. A correct touch description plus the horizontal height 1 distinguishes D from all three; equivalent decisive feature combinations are accepted.

Practice rubric (2 points)

- 1: Choice D.
- 0.5: Correct touching/double-zero behavior at x=1, not merely naming an intercept.
- 0.5: Another correct feature, provided the justification as a whole rules out A, B and C. 'Touches from below' plus any correct second feature qualifies. For a touch description without side, require HA height 1 or negative middle-branch sign; retain 0.25 for a correct but nondecisive second feature.

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11. 2 points

1. B reduces to $(x-2)(x+3)/(2(x-1))$, but retains the excluded input -1. Substitution there gives $(-3)(2)/(2(-2)) = 3/2$, so the hole is $(-1, 3/2)$.
2. The vertical asymptote is $x = 1$; division gives $y = x/2 + 1$ as the slant. These agree with the graph, as do zeros -3 and 2.
3. A fills the missing point. In C, an x+1 factor remains in the denominator after cancellation, giving a pole at -1, not a removable hole there. D reflects the curve and gives the wrong hole height sign. A common factor by itself is not enough to decide the entire graph.

Practice rubric (2 points)

- 1: Choice B.
- 0.5: Hole coordinates (-1,3/2); 0.25 for the correct input/cancellation mechanism alone. Accept a reasonable graph estimate of the height supported by the correct selected rule.
- 0.5: Another distinguishing feature, such as VA x=1, with reasoning that does not treat every common factor as automatically a hole. Together the cited features must be consistent with the whole graph.

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12. 2 points

1. The reference f has positive even leading term, so both ends go up. For $m = 8$, division by the leading x^2 gives leading behavior $4x^6$, also both ends up.
2. For $m = 7$, the leading behavior is $4x^5$, giving opposite ends. For $m = 6$, combine the tied numerator terms to get $-3x^4$ after division; for $m = 2$, the leading behavior is $-7x^4$. Both of those have ends down. Matching directions does not require matching the exact degree.

Practice rubric (2 points)

- 1: Choice A.
- 1: Correct leading sign and parity after accounting for the denominator; accept concise equivalent reasoning, not a required rejection of every other option.

13. 4 points

1. Near 1, the outputs remain finite and approach about 2.11 from both sides, supporting the hole at $x = 1$ under the stated assumptions.
2. Near -2, positive outputs grow rapidly in magnitude from both sides. Thus the likely vertical line is $x = -2$, with $H(x) \to +\infty$ as $x \to -2^-$ and as $x \to -2^+$. A vertical asymptote need not have opposite signs on its two sides.
3. One parent-only generating rule is $H(x) = 2 + 1/(x + 2)^2$, additionally excluding 1. Its missing height is $19/9$; students are not asked to find the rule or prove an exact limit from finite rounded samples.

Practice rubric (4 points)

- 1: Hole at 1.
- 1: Finite-height table evidence near 1; do not require exact 19/9.
- 1: Vertical asymptote at -2.
- 1: Growing-magnitude evidence 0.5; correct positive-infinity behavior from both sides 0.5 (0.25 each).